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Models Gone Wild: Taming AI Chaos with OpenShift AI

Jeff Brassard - Solutions Architect

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The AI Goldrush

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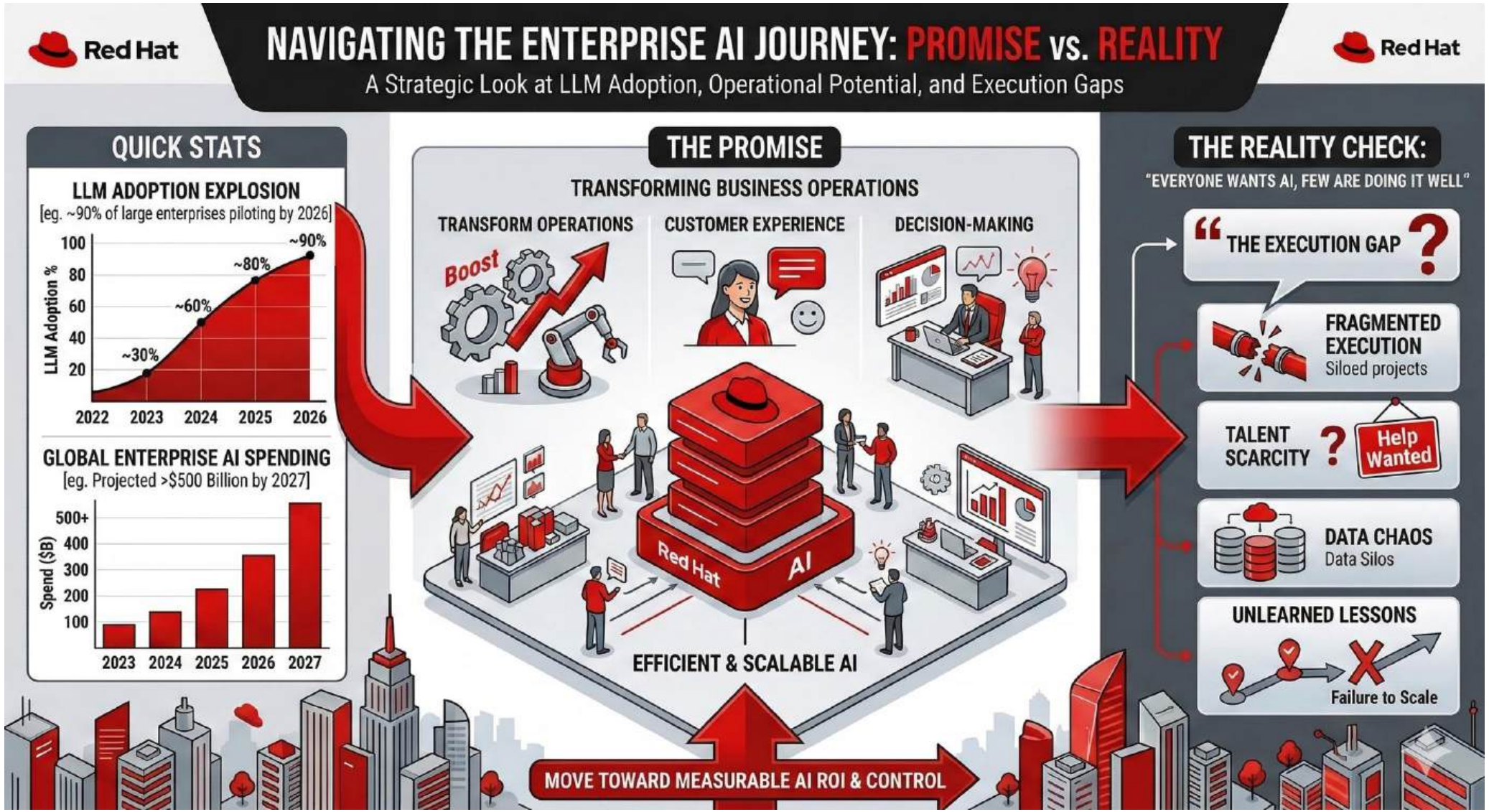
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NAVIGATING THE ENTERPRISE AI JOURNEY: PROMISE vs. REALITY
A Strategic Look at LLM Adoption, Operational Potential, and Execution Gaps
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QUICK STATS
LLM ADOPTION EXPLOSION
[eg. ~90% of large enterprises piloting by 2026]
LLM Adoption %
100
80
60
40
20
~30%
~60%
~80%
~90%
2022
2023
2024
2025
2026
GLOBAL ENTERPRISE AI SPENDING
[eg. Projected >$500 Billion by 2027]
Spend ($B)
500+
400
300
200
100
2023
2024
2025
2026
2027
THE PROMISE
TRANSFORMING BUSINESS OPERATIONS
TRANSFORM OPERATIONS
CUSTOMER EXPERIENCE
DECISION-MAKING
Boost
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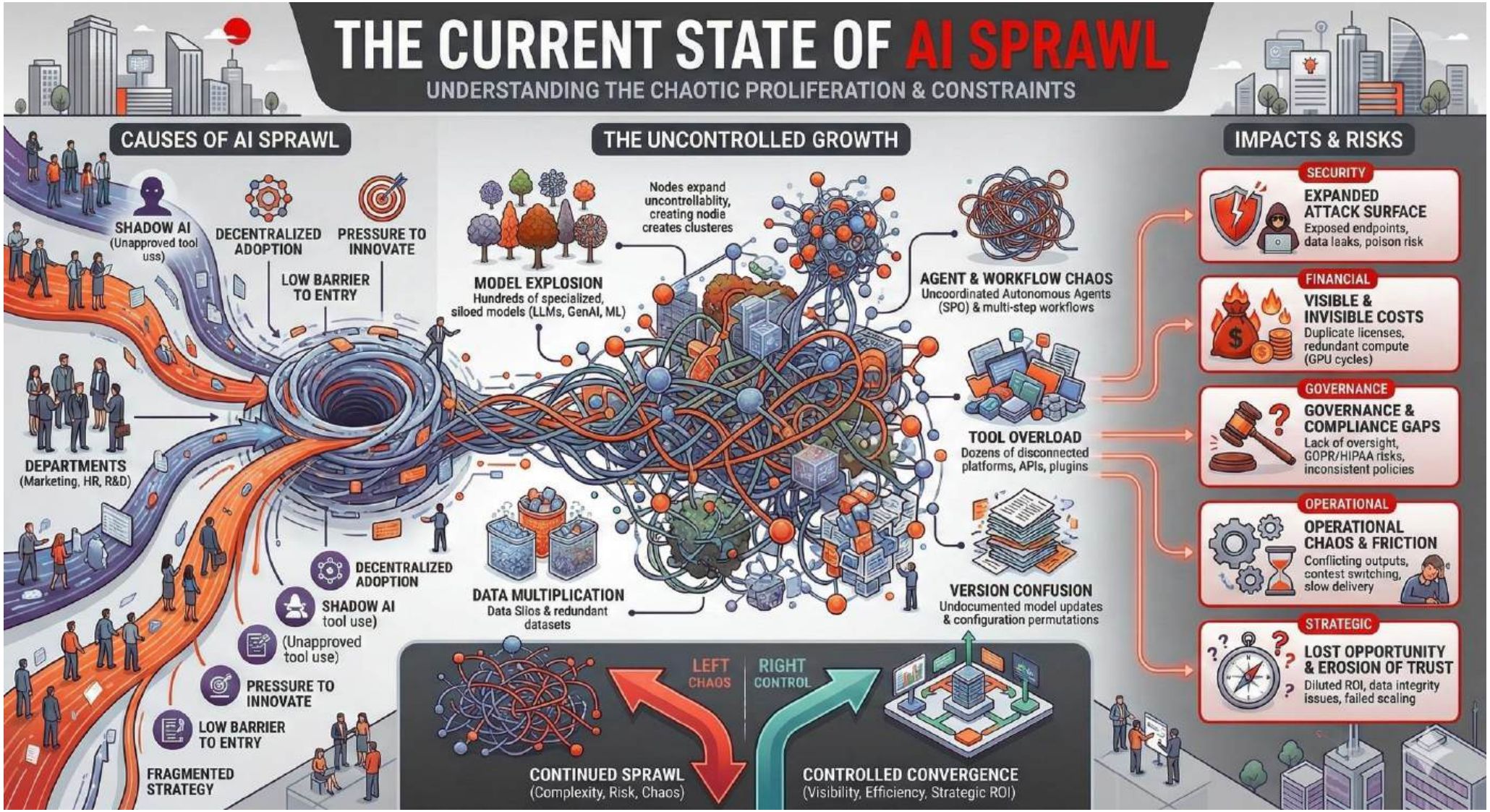
AI
EFFICIENT & SCALABLE AI
MOVE TOWARD MEASURABLE AI ROI & CONTROL
THE REALITY CHECK:
"EVERYONE WANTS AI, FEW ARE DOING IT WELL"
"THE EXECUTION GAP ?
FRAGMENTED EXECUTION
Siloed projects
TALENT SCARCITY ?
Help Wanted
DATA CHAOS
Data Silos
UNLEARNED LESSONS
Failure to Scale


THE CURRENT STATE OF AI SPRAWL
UNDERSTANDING THE CHAOTIC PROLIFERATION & CONSTRAINTS
CAUSES OF AI SPRAWL
SHADOW AI (Unapproved tool uss)
DECENTRALIZED ADOPTION
PRESSURE TO INNOVATE
LOW BARRIER TO ENTRY
DEPARTMENTS (Marketing, HR, R&D)
DECENTRALIZED ADOPTION
SHADOW AI tool use)
(Unapproved tool use)
PRESSURE TO INNOVATE
LOW BARRIER TO ENTRY
FRAGMENTED STRATEGY
THE UNCONTROLLED GROWTH
Nodes expand uncontrollablity, creating nodie creates clusteres
MODEL EXPLOSION
Hundreds of specialized, siloed models (LLMs, GenAI, ML)
AGENT & WORKFLOW CHAOS
Uncoordinated Autonomous Agents (SPO) & multi-step workflows
TOOL OVERLOAD
Dozens of disconnected platforms, APIs, plugins
DATA MULTIPLICATION
Data Silos & redundant datasets
VERSION CONFUSION
Undocumented model updates & configuration permutations
LEFT CHAOS
RIGHT CONTROL
CONTINUED SPRAWL
(Complexity, Risk, Chaos)
CONTROLLED CONVERGENCE
(Visibility, Efficiency, Strategic ROI)
IMPACTS & RISKS
SECURITY
EXPANDED ATTACK SURFACE
Exposed endpoints, data leaks, poison risk
FINANCIAL
VISIBLE & INVISIBLE COSTS
Duplicate licenses, redundant compute (GPU cycles)
GOVERNANCE
GOVERNANCE & COMPLIANCE GAPS
Lack of oversight, GDPR/HIPAA risks, inconsistent policies
OPERATIONAL
OPERATIONAL CHAOS & FRICTION
Conflicting outputs, contest switching, slow delivery
STRATEGIC
LOST OPPORTUNITY & EROSION OF TRUST
Diluted ROI, data integrity issues, failed scaling

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Why AI Is Hard

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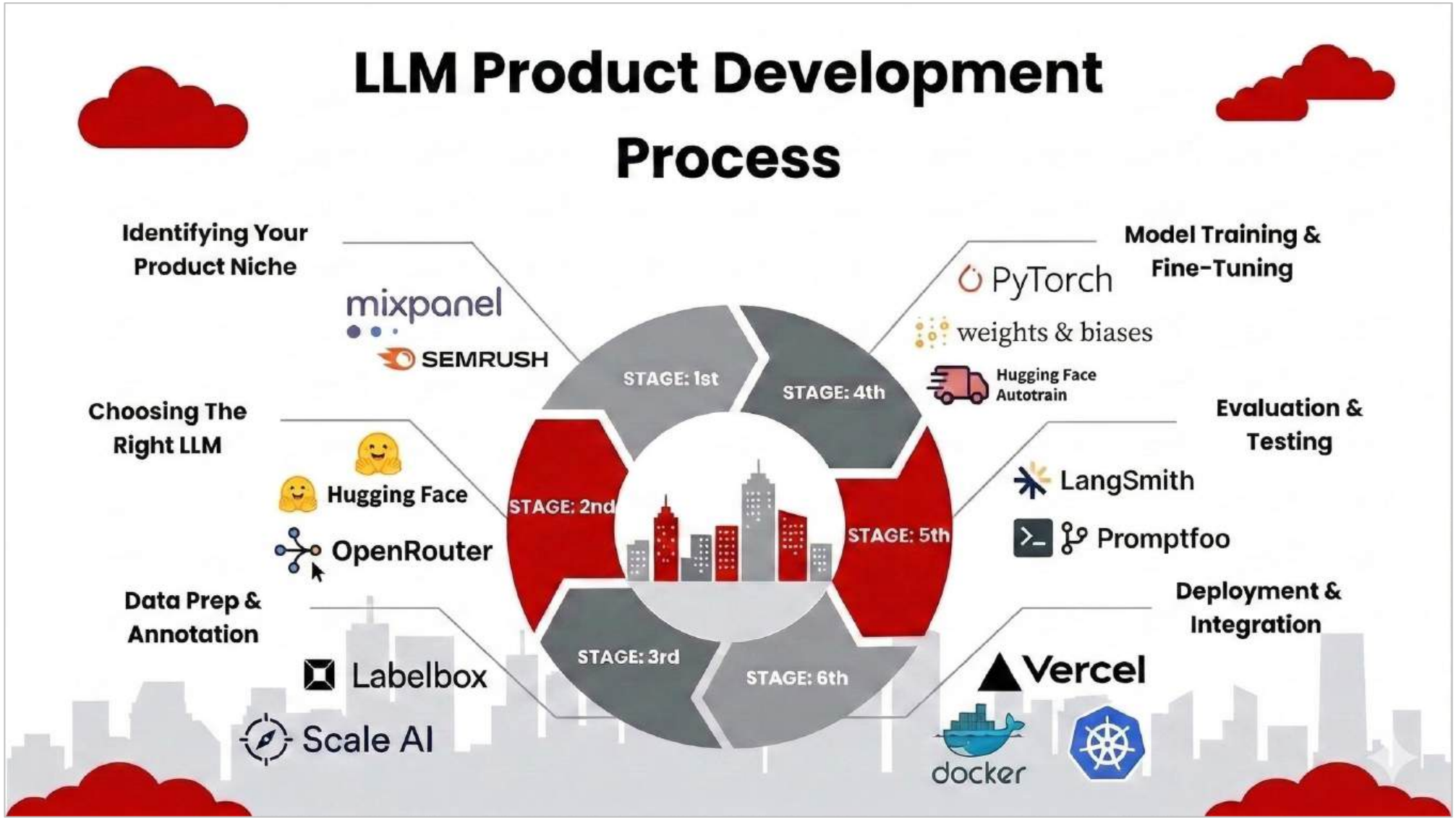
LLM Product Development Process
Identifying Your Product Niche
mixpanel
SEMRUSH
Choosing The Right LLM
Hugging Face
OpenRouter
Data Prep & Annotation
Labelbox
Scale AI
STAGE: 1st
STAGE: 2nd
STAGE: 3rd
STAGE: 4th
STAGE: 5th
STAGE: 6th
Model Training & Fine-Tuning
PyTorch
weights & biases
Hugging Face Autotrain
Evaluation & Testing
LangSmith
Promptfoo
Deployment & Integration
Vercel
docker

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The “Production Gap” Problem

The Most AI Projects never reach production / deliver on “bottom-line impact”

"80-85% of AI projects fail to reach production" (RAND/Gartner)”

- ✓ RAND Corporation (2024), Gartner (2024)

"Despite 78% of organizations using AI, over 80% see no bottom-line impact" (McKinsey)

- ✓ McKinsey State of AI Report (2024)

Challenges:

- ✓ Infrastructure Challenges
- ✓ GPU availability and costs spiraling
- ✓ Security and compliance requirements
- ✓ Lack of standardization across environments

The Multi-Cloud Dilemma

AWS
(Amazon Web Services)

AWS
(Amazon Web Services)

GCP
(Google Cloud Platform)

GCP
(Google Cloud Platform)

Azure
(Microsoft Azure)

Azure
(Microsoft Azure)

On-Premises Infrastructure

Secure Private Data Centers (On-Premises)

- ✓ Your data lives everywhere
- ✓ Vendor lock-in
- ✓ Migration nightmares
- ✓ Inconsistent tooling and skills required
- ✓ $$$

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Enter OpenShift AI

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What if AI Could Be… Boring?

- ✓ Boring = Reliable, Repeatable, Production-Ready
- ✓ OpenShift AI makes AI Infrastructure predictable
- ✓ Built on proven enterprise Kubernetes
- ✓ Certified Model Catalog
- ✓ Best in-class AI inferencing (vLLM and llm-d)

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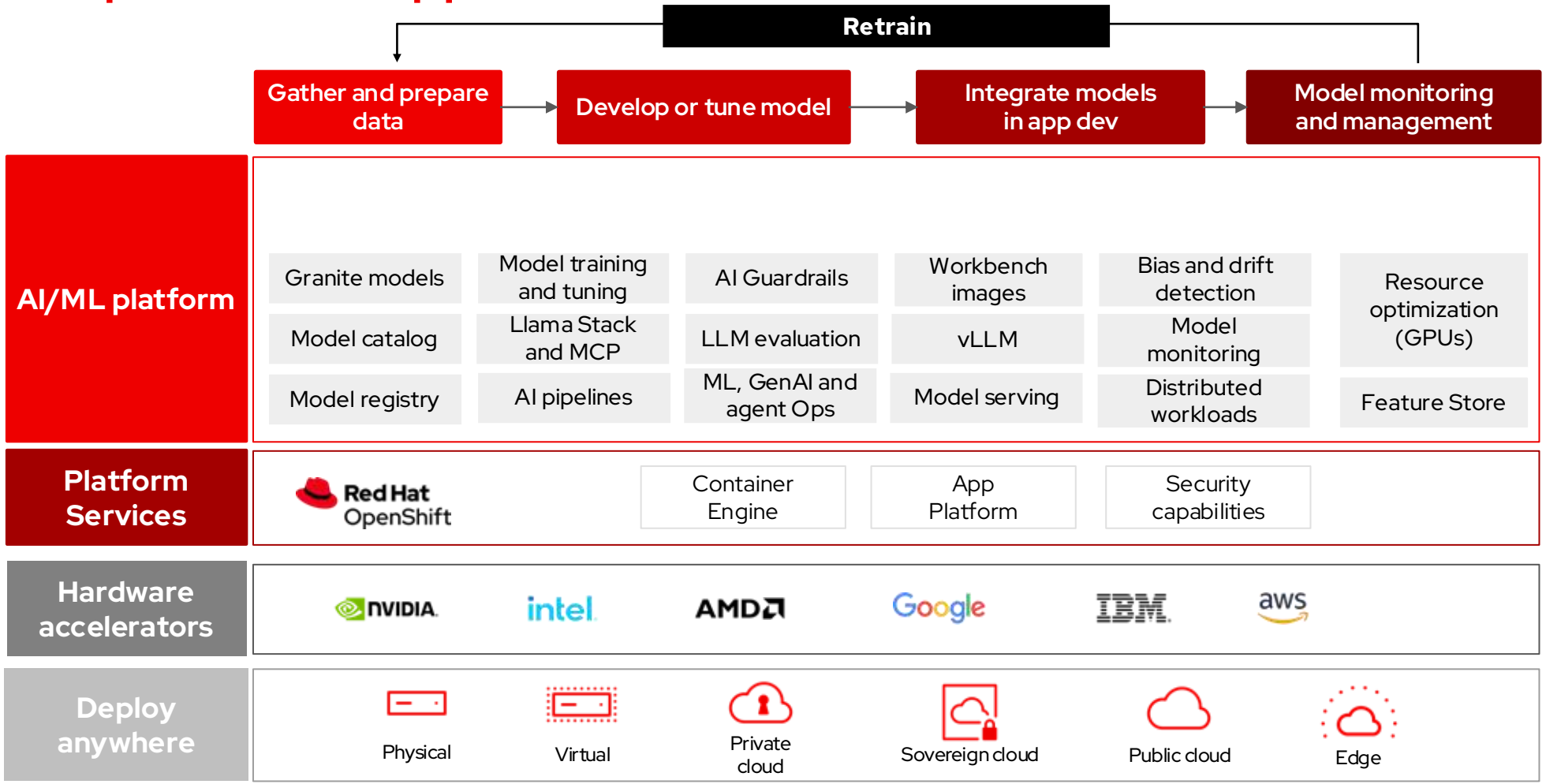
The OpenShift AI Approach



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Deploy Anywhere, Actually

Hardware Accelerators

NVIDIA
GPU

AMD
Instinct

intel.
Gaudi

Google
TPU

IBM
AIU

aws
Neuron

OEM Servers

DELL

Lenovo

cisco

Hewlett Packard Enterprise

Cloud Environments

IBM Cloud

aws

Google Cloud

Azure

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From Chaos To Control

Increased efficiency

Reduce development and deployment costs with access to optimized open source models

Simplified experience

Enable developers, data scientists and domain experts to tailor models more efficiently

Flexibility to deploy anywhere

Mitigate risks, reduce costs, and scale your AI deployments across the hybrid cloud

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Why OpenShift AI

Help address AI challenges: cost, efficiency, and flexibility

CUSTOMER REALITIES

- **High costs** of running large generative AI models aligned with enterprise data and use cases
- **Fragmentation** in tooling & approaches across AI lifecycle creates friction, delays, risk, inconsistency
- The **need for flexible deployment** across the hybrid cloud

RED HAT VALUE ADD

- **Lower Costs:** utilize smaller, efficient language models where possible
- **Governance & Accessibility:** Unify full AI/ML lifecycle, from lab to production
- **Deploy Anywhere / any workload:** any model, any accelerator, across any cloud

BUSINESS IMPACT

- **Resource Control**: Optimize models and reduce overhead
- **Accelerated Innovation**: Simplify adoption with customized fit for use solutions
- **Future Proof:** Solving immediate objectives, staying ahead of curve without re-platforming

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Thank you

Red Hat is the world's leading provider of enterprise open source software solutions. Award-winning support, training, and consulting services make Red Hat a trusted adviser to the Fortune 500.

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